

Specular Grassmann-Polarity Index Method

Application to Ebola Virus - Version 12.0

USER MANUAL

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Abstract

This document serves as the complete user’s manual and conceptual framework for SGPMAN (version 12.0), a computational platform for protein analysis and peptide design, applied specifically to the Ebola Virus (EBOV). The system is built upon four fundamental pillars: (1) the Polarity Index Method (PIM), which represents proteins as 16-dimensional vectors; (2) Grassmann Algebra, which serves as the mathematical engine for comparing and transforming polarity profiles; (3) Transfer Learning with ESM2 embeddings for enriched sequence representation; and (4) Multi-Objective Bayesian Optimization for peptide design.

New Features in SGPMAN (version 12.0):

- **Refinements in Clifford Algebra:** The complete geometric product $v * w = v \cdot w + v \wedge w$ has been further refined for numerical stability and efficiency, allowing for a richer decomposition of the relationship between two PIM vectors.
- **Enhanced Rotor Similarity:** The rotor concept for optimal alignment of PIM vectors has been improved with better numerical stability, providing a more robust measure of similarity and alignment.
- **Rényi Entropy Suite:** The full family of Rényi entropies ($\alpha = 1, 2, 3, \infty$) is now integrated into the core analysis, providing a multi-faceted view of the diversity of polarity transitions. The implementation is numerically stable and includes Shannon entropy as a special case.
- **Expanded Chemical Profiling:** A comprehensive chemical profile is now generated for each target protein, including over 24 properties such as Net Charge, Charge Density, GRAVY, Hydrophobicity Score, Solubility, ΔG , Tm, Stability Score, and Hotspot analysis (hydrophobic/charge-rich regions).
- **Streamlined Workflow for EBOV:** The system has been specifically configured and tested for the analysis of Ebola Virus (species: Zaire, Sudan, Reston, Bundibugyo, Tai Forest, Bombali). This includes target-specific ranges and configurations to guide the peptide design process.
- **Comprehensive Report Generation:** The system generates a multidisciplinary summary table, narrative summaries tailored for different scientific profiles (Executive, Biochemist, Chemist, Analytical Chemist, Physicochemist, Bioinformatician), and detailed chemical profiles for each target. This facilitates a holistic understanding of the protein’s characteristics.

Version Information: This manual corresponds to SGPMAN (version 12.0) (file: SGPMAN212.0v.py).

Execution Log Summary (September 2026 for EBOV Analysis):

- Total sequences processed: 1,832,319
- Total valid PIMs: 1,832,319 (100.00% validity)
- Total rejected: 0
- Total bytes processed: 0.57 GB

- Processing time: 6 minutes 37 seconds
- Average speed: 4,613 sequences per second
- Peak memory usage (streaming): ~8 MB
- Peak memory usage (analysis): ~500 MB
- Hash index built: 1,792 unique buckets from 1,888 proteins
- *config_EBOLA.json* loaded and validated successfully
- Reference group used: EBOLA_ZAIRE

Part I

Understanding SGPMAN (version 12.0): The Program's Vision and Architecture

INTRODUCTION: WHY SGPMMAIN (VERSION 12.0) EXISTS

1.1 The Fundamental Problem

Proteins are the molecular machines of life. They catalyze reactions, transport molecules, recognize pathogens, and transmit signals. Understanding how proteins work—and how to design molecules that interact with them—is one of the most important challenges in modern biology and medicine.

For decades, the dominant approach to understanding proteins has been through sequence analysis. We compare amino acid sequences, look for conserved motifs, and infer function from homology. This approach has been enormously successful, but it has a fundamental limitation: it treats proteins as strings of letters rather than as physical entities with specific physicochemical properties.

The insight behind SGPMMAIN (version 12.0) is that protein function is not determined by the exact sequence of amino acids, but by the spatial and temporal organization of polarity transitions along the polypeptide chain. Two proteins with completely different sequences can have very similar functions if they share similar polarity patterns. Conversely, proteins with similar sequences can have different functions if their polarity patterns diverge.

This insight has profound practical implications. It enables the design of therapeutic peptides not by copying natural sequences, but by understanding the polarity patterns that underlie protein function. It enables the identification of new drug targets by finding proteins that share polarity patterns with known targets. And it enables us to move beyond the limitations of sequence alignment to a more fundamental understanding of protein function.

1.2 The Four Pillars of SGPMMAIN (version 12.0)

SGPMMAIN (version 12.0) is built on four pillars that work together to transform raw sequence data into actionable biochemical insights:

Pillar 1: The Polarity Index Method (PIM)

The Polarity Index Method is the foundational representation layer. It transforms the complex, high-dimensional space of amino acid sequences into a compact 16-dimensional vector space. Each dimension corresponds to a specific polarity transition: from positive to negative charge, from polar to nonpolar, and so on.

This transformation is mathematically rigorous yet biologically interpretable. Each transition directly corresponds to a specific physicochemical interaction that can be measured and

validated experimentally. The resulting 16-dimensional vector is a fingerprint of the protein’s polarity organization that is independent of sequence length and robust to minor sequence variations.

Pillar 2: Grassmann Algebra with Enhanced Mathematical Framework

Grassmann Algebra is the mathematical engine that operates on the PIM vectors. Version 12.0 builds upon the 18 improvements from version 11.0, refining them for efficiency and robustness.

The key operations at this level include:

- **Complete and Refined Geometric Product:** The fundamental operation remains $v * w = v \cdot w + v \wedge w$, but now with a more stable and efficient implementation.
- **Enhanced Rotor Similarity:** The rotor concept for aligning PIM vectors has been refined for greater numerical robustness.
- **Optimal Geodesic:** The optimal trajectory between two points in Grassmann space is computed using SVD decomposition.
- **Multilevel Analysis:** Grassmann analysis is performed at multiple levels ($k = 1, 2, 3$), capturing geometric structure at different scales.

Pillar 3: Transfer Learning with ESM2 and Advanced Metrics

Protein language models like ESM2 have revolutionized computational biology. Trained on millions of protein sequences, these models learn representations that capture evolutionary, structural, and functional information.

In SGPMAN (version 12.0), ESM2 embeddings are combined with PIM vectors and physico-chemical properties to create enriched feature vectors. Version 12.0 enhances this pillar with a refined suite of metrics:

- **Rényi Entropy:** A complete family of entropy measures including Shannon ($\alpha = 1$), collision ($\alpha = 2$) and min-entropy ($\alpha \rightarrow \infty$).
- **Bhattacharyya Distance:** A metric complementary to Hellinger.
- **Entropic Wasserstein (Sinkhorn):** Optimal transport with entropic regularization.
- **Functional PCA:** Treatment of PIM vectors as continuous functions using B-splines.

Pillar 4: Hybrid Configuration with *config_EBOLA.json* and Multi-Objective Optimization

Version 12.0 continues to use a hybrid configuration with *config_EBOLA.json* and multi-objective optimization. This pillar includes:

- ***config_EBOLA.json* with Fallback:** Structured external configuration with default values.
- **Target-Specific Ranges:** Each virus or protein has its own metric ranges.
- **Dynamic Metric Weights:** Metric weights can be adjusted per target.
- **Vectorized NSGA-II:** The NSGA-II multi-objective optimization algorithm is fully vectorized.

1.3 The Three-Level Architecture

SGPMAIN (version 12.0) operates through a carefully designed three-level architecture that transforms raw sequence data into actionable biochemical insights. Each level builds upon the previous one, creating a coherent pipeline from sequence to function.

Level 1: PIM — The Foundation

At the most fundamental level, SGPMAIN (version 12.0) implements the Polarity Index Method. This level takes as input the amino acid sequence of a protein and produces as output a 16-dimensional vector. The transformation proceeds through three steps:

First, each amino acid in the sequence is classified into one of four polarity categories: positively charged (P^+ : H, K, R), negatively charged (P^- : D, E), neutral polar (N: C, G, N, Q, S, T, Y), and nonpolar hydrophobic (NP: A, F, I, L, M, P, V, W). This classification captures the essential electrostatic and hydrogen-bonding characteristics of each residue.

Second, the sequence is scanned to count the frequency of each possible transition between polarity categories. With four categories, there are 16 possible transitions, and these counts form the components of the PIM vector.

Third, the vector is normalized so that all components sum to unity, producing a probability distribution that is independent of sequence length. This normalization ensures that proteins of different lengths can be compared directly.

The result is a compact, length-independent representation of the protein's polarity organization that captures the essential physicochemical information needed for functional analysis.

Level 2: Grassmann Algebra and Transfer Learning — The Analytical Engine

At the intermediate level, SGPMAN (version 12.0) applies Grassmann Algebra and Transfer Learning to operate on the PIM vectors produced by Level 1. Grassmann algebra provides a natural mathematical framework for working with vector subspaces and their geometric relationships. ESM2 provides enriched evolutionary and functional context.

The key operations at this level include:

- **Complete Geometric Product:** Computes the decomposition of the product $v * w$ into scalar, vector, and bivector parts.
- **Rotor Similarity:** Computes the optimal rotation that aligns one PIM vector with another.
- **Rényi Entropy:** Characterizes the diversity of polarity transitions.
- **Entropic Wasserstein:** Computes the optimal cost of transforming one polarity distribution into another.
- **FPCA:** Identifies the main modes of functional variation in PIM profiles.

Level 3: Derived Properties, PIDP, and Peptide Design — The Biochemical Outputs

At the highest level, SGPMAN (version 12.0) transforms the abstract vector space of PIM vectors into concrete biochemical properties that are directly interpretable by experimental scientists. This level comprises four complementary modules:

- **Chemical Properties Module:** Derives over 24 physicochemical properties from the PIM vector, including net charge, GRAVY, solubility, ΔG , T_m , stability score, and hotspot analysis.
- **Protein Intrinsic Disorder Prediction (PIDP) Module:** Provides critical structural insights using two independent methods: metapredict (machine learning-based) and AIUPred (biophysical model-based).
- **Multi-Objective Predictor Module:** Predicts multiple properties of antiviral peptides: antiviral activity, cytotoxicity, stability, and selectivity index.
- **Peptide Design Module:** Designs optimized peptides targeting specific proteins. It identifies critical polarity transitions, designs peptide sequences that mimic these transitions, uses Gaussian Process optimization, and evaluates drug-likeness.

1.4 The Role of *config_EBOLA.json* in the Architecture

The *config_EBOLA.json* file is a critical addition in version 12.0 that fundamentally changes how SGPMAIN operates. It sits at the boundary between Level 2 and Level 3, providing target-specific parameters that guide both the analytical engine and the biochemical outputs.

In previous versions, all parameters were hardcoded. To change the metric weights, users had to modify the source code. To add a new virus, they had to edit the MAIN_GROUP list. To adjust the batch size, they had to find the right variable in the code and change it. This approach made the program inflexible, hard to maintain, and difficult to adapt to new targets.

The *config_EBOLA.json* file is a structured JSON document that contains all the configurable parameters of SGPMAIN (version 12.0). It is divided into sections that mirror the architecture of the program:

- **Metadata:** Version information, generation date, and source information.
- **Metric Weights:** The weights of the 13 metrics used in peptide design.
- **Target Ranges:** For each virus or protein target, the acceptable ranges for each metric (PIM similarity, Rényi entropy, Grassmann distance, etc.).
- **Classification Thresholds:** The thresholds for excellent, good, moderate, and poor drug candidates.
- **Known Inhibitors:** For each virus, the known inhibitors with their IC50 and Ki values.
- **Virus-Specific Parameters:** UniProt IDs, PDB structures, domains, key residues, serotypes, and clinical considerations.
- **Peptide Design Parameters:** Preferred amino acids, length ranges, secondary structure preferences, and known resistance mutations.
- **Base Peptide Sequence:** The known antiviral peptide to be used as starting point for optimization.
- **Processing Parameters:** Batch size, sample size, bootstrap iterations, and worker count.
- **Data Paths:** Paths to external files like *chembl_uniprot.txt* and *apd_natural.fasta*.

Target-Specific Optimization

The most powerful feature of *config_EBOLA.json* is target-specific optimization. Each virus or protein target can have its own set of parameters, allowing the system to be optimized for

the specific characteristics of that target. In this study, the system is configured to target the Ebola Virus glycoprotein (GP), with specific parameters for different species (Zaire, Sudan, Reston, Bundibugyo, Tai Forest, Bombali).

1.5 What SGPMAN (version 12.0) Does Not Do

Understanding what SGPMAN (version 12.0) does not do is as important as understanding what it does.

Not a structure prediction tool: SGPMAN does not predict the three-dimensional structure of proteins. Instead, it predicts the polarity organization of proteins, which is a different and complementary type of information.

Not a docking tool: SGPMAN does not predict how two proteins will bind to each other. Instead, it predicts whether two proteins are likely to share similar functions based on their polarity profiles.

Not a drug design tool: SGPMAN does not design drugs in the traditional sense. Instead, it designs peptides that mimic the polarity patterns of target proteins, which can then be synthesized and tested as potential therapeutics.

Not a substitute for experiments: The predictions and designs produced by the system are hypotheses that need to be validated experimentally.

THE COMPLETE WORKFLOW: FROM SEQUENCE TO PEPTIDE

This chapter describes the complete workflow of SGPMAN (version 12.0), from loading protein sequences to designing therapeutic peptides. The workflow is a pipeline that transforms raw data into actionable insights.

2.1 Overview of the Workflow

The SGPMAN (version 12.0) workflow consists of nine stages, each building on the previous one:

1. **Data Loading:** The system reads protein sequences from FASTA files, one group at a time.
2. **PIM Computation:** For each protein sequence, the system computes the 16-dimensional PIM vector.
3. **Group Statistics:** For each group of proteins, the system computes statistics including the centroid (mean PIM vector), covariance matrix, and various geometric measures.
4. **Comparison and Similarity:** The system compares groups and individual proteins using the wedge similarity and other geometric measures.
5. **Chemical Profiling:** For each group, the system derives chemical properties from the PIM vector.
6. **PIDP Analysis:** The system predicts intrinsic disorder for each target protein using metapredict and AIUPred.
7. **Therapeutic Profiling:** The system designs therapeutic peptides targeting the selected protein.
8. **Peptide Design Engine:** The system uses Bayesian optimization to generate optimized peptide candidates.
9. **Drug-Likeness Filtering:** The system evaluates peptide candidates against multiple drug-likeness criteria.

2.2 Stage 1: Data Loading

The data loading stage is the entry point of the workflow. The system reads FASTA files from the working directory, each containing protein sequences for a specific group.

Input: FASTA files with the naming pattern `{group_name}.unico.dat0`.

Processing: The system streams each file, reading sequences one at a time. This streaming approach is essential for handling large files.

Algorithm Description:

1. For each file in the working directory matching the pattern `*.unico.dat0`, extract the group name.
2. Open the file and read it line by line.
3. For each sequence, parse the FASTA header and the sequence string.
4. Validate the sequence: check that it contains only standard amino acid letters (A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y).
5. Store up to `max_stored_proteins` sequences per group using reservoir sampling.
6. If the total number of stored sequences exceeds `max_stored_proteins`, perform random replacement.
7. Continue until the end of the file is reached.

Output: For each group, the system stores a sample of up to 200 protein sequences (configurable via `config_EBOLA.json`). The sample is maintained using reservoir sampling.

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- Total rejected: 0
- Validity rate: 100.00%
- Total bytes processed: 0.57 GB
- Processing time: 6 minutes 37 seconds
- Average speed: 4,613 sequences per second
- Peak memory usage (streaming): ~8 MB
- Hash index buckets: 1,792

2.3 Stage 2: PIM Computation

The PIM computation stage transforms each protein sequence into a 16-dimensional PIM vector.

Input: Protein sequence (as a string of amino acids).

Processing: The system classifies each amino acid into one of four polarity categories, counts the frequency of each polarity transition, and normalizes the resulting vector.

Algorithm Description:

1. Initialize a 16-element vector v with zeros.
2. For each pair of consecutive amino acids (a_i, a_{i+1}) in the sequence:
 - a) Determine the polarity class of a_i : $c_i = \phi(a_i)$.
 - b) Determine the polarity class of a_{i+1} : $c_{i+1} = \phi(a_{i+1})$.
 - c) Map the ordered pair (c_i, c_{i+1}) to an index j in the set $\{0, \dots, 15\}$.
 - d) Increment v_j by 1.
3. Normalize v so that $\sum_i v_i = 1$:

$$v_i \leftarrow \frac{v_i}{\sum_k v_k} \quad (2.1)$$

Output: A 16-dimensional PIM vector where each component is a probability (the sum of all components is 1).

Polarity Classification

The system uses the following polarity classification:

- **P⁺ (Positively Charged):** H, K, R
- **P⁻ (Negatively Charged):** D, E
- **N (Neutral Polar):** C, G, N, Q, S, T, Y
- **NP (Nonpolar):** A, F, I, L, M, P, V, W

2.4 Stage 3: Group Statistics

The group statistics stage computes summary statistics for each group of proteins.

Input: PIM vectors for all proteins in a group.

Processing: The system computes the centroid (mean PIM vector), covariance matrix, standard deviation, and various geometric measures, including Rényi entropy, Bhattacharyya distance, entropic Wasserstein, FPCA, mutual information, and total correlation.

Algorithm Description:

1. Compute the centroid (mean) of the PIM vectors:

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i \quad (2.2)$$

2. Compute the covariance matrix:

$$\Sigma = \frac{1}{n-1} \sum_{i=1}^n (v_i - \bar{v})(v_i - \bar{v})^T \quad (2.3)$$

3. Compute the standard deviation vector: $\sigma = \sqrt{\text{diag}(\Sigma)}$.

4. Compute the Shannon entropy of the centroid:

$$H(\bar{v}) = - \sum_i \bar{v}_i \log_2(\bar{v}_i + \epsilon) \quad (2.4)$$

5. Compute the Rényi entropy for $\alpha = 2$ and $\alpha = 3$:

$$H_\alpha(\bar{v}) = \frac{1}{1-\alpha} \log_2 \left(\sum_i \bar{v}_i^\alpha \right) \quad (2.5)$$

6. Compute the Gini coefficient of the centroid:

$$G(\bar{v}) = 1 - \frac{2}{n} \sum_{i=1}^n \frac{\sum_{j \leq i} \bar{v}_{(j)}}{\sum_j \bar{v}_j} \quad (2.6)$$

where $\bar{v}_{(1)} \leq \bar{v}_{(2)} \leq \dots \leq \bar{v}_{(n)}$.

7. Compute the fractal dimension of the centroid using box-counting.
8. Compute the Wasserstein distance between the centroid and a uniform distribution:

$$W_1(\bar{v}, U) = \sum_{i=1}^n |\bar{v}_i - 1/n| \quad (2.7)$$

9. Compute the structural complexity index:

$$C(\bar{v}) = \frac{H(\bar{v})}{\log_2(n)} \quad (2.8)$$

10. Compute the functional modularity index using spectral clustering.

11. Compute Moran's I for the centroid:

$$I(\bar{v}) = \frac{n}{S_0} \frac{\sum_i \sum_j w_{ij} (\bar{v}_i - \bar{v})(\bar{v}_j - \bar{v})}{\sum_i (\bar{v}_i - \bar{v})^2} \quad (2.9)$$

where w_{ij} is a spatial weight matrix (based on transition adjacency).

12. Compute the total correlation:

$$TC(\bar{v}) = \sum_i \bar{v}_i \log_2 \left(\frac{\bar{v}_i}{1/n} \right) \quad (2.10)$$

Output: A GroupStatistics object containing all the computed statistics.

The Centroid

The centroid of a group is the mean PIM vector of all proteins in the group. It represents the "average" polarity profile of the group.

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i \quad (2.11)$$

where n is the number of proteins in the group.

2.5 Stage 4: Comparison and Similarity

The comparison stage measures the similarity between groups and individual proteins.

Input: PIM vectors and group statistics.

Processing: The system computes the wedge similarity between the target and each group, as well as between individual proteins.

Algorithm Description:

1. For each pair of PIM vectors (v, w) , compute the wedge similarity:

$$\text{WedgeSim}(v, w) = 1 - \frac{\|\omega_{\mathcal{K}}(v, w)\|_2}{\|v\|_2 \|w\|_2} \quad (2.12)$$

where $\omega_{\mathcal{K}}$ is the corrected wedge product.

2. Compute the rotor similarity:

$$R(v, w) = \langle v, w \rangle + \|v \wedge w\|^2 \quad (2.13)$$

This represents the optimal rotation aligning v to w .

3. Compute the Grassmann projection distance:

$$d_{Gr}(v, w) = \frac{1}{\sqrt{2}} \left\| \frac{vv^T}{\|v\|^2} - \frac{ww^T}{\|w\|^2} \right\|_F \quad (2.14)$$

4. Compute the Hodge complementarity:

$$H(v, w) = \| \star (v \wedge w) \|_2 \quad (2.15)$$

where $\star$ is the Hodge star operator.

Output: Similarity scores, rankings, angle analysis, decomposition analysis, and comparison matrices.

The Wedge Similarity

The wedge similarity is the primary measure of similarity between two PIM vectors. It is computed using the corrected wedge product.

$$\text{WedgeSim}(v, w) = 1 - \frac{\|\omega_{\mathcal{K}}(v, w)\|_2}{\|v\|_2 \|w\|_2} \quad (2.16)$$

The wedge similarity ranges from 0 to 1, where 1 indicates identical polarity profiles and 0 indicates orthogonal profiles.

2.6 Stage 5: Chemical Profiling

The chemical profiling stage derives chemical properties from the PIM vector.

Input: PIM vector for a protein or group.

Processing: The system applies a series of formulas to compute chemical properties.

Algorithm Description:

1. Compute the net charge:

$$Q = \sum_{a \in \{K, R, H\}} f_a - \sum_{a \in \{D, E\}} f_a \quad (2.17)$$

where f_a is the frequency of amino acid a in the sequence.

2. Compute the charge density:

$$\rho_Q = \frac{Q}{L} \quad (2.18)$$

where L is the sequence length.

3. Compute the GRAVY score:

$$\text{GRAVY} = \sum_{a \in \mathcal{A}} f_a \cdot \text{hydropathy}(a) \quad (2.19)$$

where hydropathy is the Kyte-Doolittle hydropathy scale.

4. Compute the solubility using an empirical model:

$$S = S_0 \cdot \exp(\alpha Q + \beta \text{GRAVY} + \gamma \cdot \text{disorder}) \quad (2.20)$$

where S_0 , α , β , γ are fitted parameters.

5. Compute the folding free energy (ΔG) using an empirical model.
6. Compute the melting temperature (T_m) using an empirical model.
7. Compute the stability score:

$$\text{Stability} = \sigma_0 + \sigma_1 \cdot Q + \sigma_2 \cdot \text{GRAVY} + \sigma_3 \cdot \text{disorder} \quad (2.21)$$

8. Identify hydrophobic patches and charge-rich regions:

- Scan the sequence with a sliding window of size w .
- For each window, compute the average hydrophobicity.
- If the average exceeds a threshold, mark the window as a hydrophobic patch.
- For charge-rich regions, compute the absolute charge density in each window.

Output: A comprehensive chemical profile with 24 properties.

Key Chemical Properties

The system computes over 24 chemical properties including: charge properties (net charge, charge density, charge balance), GRAVY (hydrophobicity), hydrophobicity score, number of hydrophobic components, hydrophobic patch components and scores, solubility, aggregation score, ΔG (folding free energy), T_m (melting temperature), stability score, hotspot analysis (hydrophobic patch, charge-rich region), membrane permeability, membrane affinity, optimal pH, salt tolerance, and buffer recommendation.

2.7 Stage 6: PIDP Analysis

The PIDP analysis stage is a critical component of the workflow. It predicts intrinsic disorder for each target protein using two independent methods: metapredict and AIUPred.

Input: Target protein sequences.

Processing: The system uses metapredict and AIUPred to predict per-residue disorder scores.

Algorithm Description (metapredict):

1. Metapredict uses a gradient-boosting machine trained on the PDB and DisProt databases.
2. For each position in the sequence, it computes a set of features:
 - Amino acid identity (one-hot encoding).
 - Predicted secondary structure (from PSIPRED or similar).
 - Amino acid properties (hydrophobicity, charge, etc.).
 - Evolutionary conservation (from PSSM or similar).
3. The model predicts a disorder score $p_i \in [0, 1]$ for each position.
4. A residue is considered disordered if $p_i > \tau$, where τ is a threshold.

Algorithm Description (AIUPred):

1. AIUPred is based on the ANCHOR2 algorithm.
2. It uses a biophysical model of binding energy.
3. For each position, it computes:

$$\text{score}_i = \frac{1}{Z} \sum_j \exp(-\Delta G_{ij}/RT) \quad (2.22)$$

where ΔG_{ij} is the binding energy between residue i and a putative partner j .

4. The score is normalized to the range $[0, 1]$.

Output: Disorder percentages at multiple thresholds (0.3, 0.4, 0.5) and mean disorder scores.

2.8 Stage 7: Therapeutic Profiling

The therapeutic profiling stage designs therapeutic peptides targeting the selected protein.

Input: Target protein PIM vector and group statistics.

Processing: The system identifies critical polarity differences between the target and host proteins, and designs a peptide sequence that mimics the target's polarity profile.

Algorithm Description:

1. Identify the target protein's PIM vector v_{target} .
2. Identify the host cell protein's PIM vector v_{host} (from a reference group).
3. Compute the difference vector:

$$\Delta = v_{\text{target}} - v_{\text{host}} \quad (2.23)$$

4. Identify the most significant differences (largest positive components of Δ).
5. For each significant difference, design a peptide sequence that mimics the target's polarity profile:
 - a) Determine the polarity categories that need to be enriched.
 - b) Search for amino acids that have high propensity for those polarity categories.
 - c) Construct a peptide sequence that maximizes the similarity to the target.
6. Evaluate the designed peptide using all 13 metrics.

Output: A therapeutic peptide with predicted activity, physicochemical properties, and comparison with known inhibitors.

Peptide Design Strategy

The system uses a weighted combination of all 13 metrics to design the peptide:

$$\text{Composite} = \sum_{m \in \mathcal{M}} w_m \cdot \Delta_m \quad (2.24)$$

where $\mathcal{M}$ is the set of metrics, w_m are the metric weights (from *config_EBOLA.json*), and Δ_m are the differences between the target and host.

2.9 Stage 8: Peptide Design Engine

The peptide design engine uses Bayesian optimization to generate optimized peptide candidates.

Input: Target sequence, training data from APD.

Processing: The system uses Gaussian Process regression with UCB acquisition to explore the peptide sequence space.

Algorithm Description:

1. Initialize the peptide sequence space $\mathcal{S}$ with the base peptide sequence.
2. For $t = 1$ to T (where T is the number of iterations):
 - a) For each candidate peptide $s \in \mathcal{S}$, compute:
 - The PIM vector v_s .
 - The activity score a_s (from the predictor).
 - The drug score d_s (from the predictor).
 - The similarity $sim_s = \text{WedgeSim}(v_s, v_{\text{target}})$.

- b) Fit a Gaussian Process to the observed scores:

$$f(s) \sim \mathcal{GP}(\mu(s), k(s, s')) \quad (2.25)$$

where k is the Matern kernel.

- c) Compute the acquisition function for each candidate:

$$\text{UCB}(s) = \mu(s) + \beta\sigma(s) \quad (2.26)$$

where β is the exploration parameter.

- d) Select the candidate with the highest acquisition value.
e) Mutate the selected sequence to generate new candidates.

3. Return the Pareto front of optimized candidates.

Output: A Pareto front of optimized peptide candidates.

2.10 Stage 9: Drug-Likeness Filtering

The drug-likeness filtering stage evaluates peptide candidates against multiple criteria.

Input: Peptide candidates and their predicted properties.

Processing: The system applies Lipinski-modified rules, Veber rules, toxicity filters, stability filters, and synthesis feasibility filters.

Algorithm Description:

1. For each candidate peptide s , compute:
 - Molecular weight (MW).
 - Number of hydrogen bond donors (HBD).
 - Number of hydrogen bond acceptors (HBA).
 - LogP (partition coefficient).
 - Number of rotatable bonds (RB).
 - Polar surface area (PSA).

2. Apply the Lipinski-modified rules:

$$\text{Lipinski}(s) = \begin{cases} 1 & \text{if } \text{MW} \leq 5000, \text{HBD} \leq 10, \text{HBA} \leq 20, \text{LogP} \leq 5 \\ 0 & \text{otherwise} \end{cases} \quad (2.27)$$

3. Apply the Veber rules:

$$\text{Veber}(s) = \begin{cases} 1 & \text{if } \text{RB} \leq 10, \text{PSA} \leq 140 \\ 0 & \text{otherwise} \end{cases} \quad (2.28)$$

4. Apply toxicity filters based on sequence motifs.
5. Apply stability filters:

$$\text{StabilityFilter}(s) = \begin{cases} 1 & \text{if predicted stability} \geq \text{threshold} \\ 0 & \text{otherwise} \end{cases} \quad (2.29)$$

6. Apply synthesis feasibility filters.

Output: A drug-likeness score and recommendation for each candidate.

The Total Drug-Likeness Score

The total drug-likeness score is a weighted sum:

$$\mathcal{D}(S) = \sum_{f \in \mathcal{F}} w_f \cdot \text{Filter}_f(S) \quad (2.30)$$

where $\mathcal{F}$ is the set of filters and w_f are their weights.

Part II

Mathematical Foundations and Metrics

THE POLARITY INDEX METHOD (PIM): MATHEMATICAL FOUNDATIONS

This chapter presents the mathematical foundations of the Polarity Index Method (PIM), including the definition of amino acid polarity, the construction of the PIM vector, the 16 polarity interactions, biological weights, the metric signature, and key functions for PIM computation.

3.1 Introduction to the Mathematical Framework

The Polarity Index Method (PIM) is built on a rigorous mathematical foundation that transforms the complex, discrete space of amino acid sequences into a continuous vector space. The key insight is that the physicochemical properties of proteins can be captured by the frequencies of transitions between polarity categories, which are independent of the specific amino acid identities.

Definition 3.1.1 (Amino Acid Polarity). *Let $\mathcal{A} = \{A_1, \dots, A_{20}\}$ be the set of the 20 standard amino acids. We define the polarity function $\phi : \mathcal{A} \rightarrow \{P^+, P^-, N, NP\}$ as:*

$$\phi(A) = \begin{cases} P^+ & \text{if } A \in \{H, K, R\} \\ P^- & \text{if } A \in \{D, E\} \\ N & \text{if } A \in \{C, G, N, Q, S, T, Y\} \\ NP & \text{if } A \in \{A, F, I, L, M, P, V, W\} \end{cases} \quad (3.1)$$

Definition 3.1.2 (PIM Vector). *For a protein sequence $S = s_1 s_2 \dots s_L$ of length L , we define the PIM vector $v \in \mathbb{R}^{16}$ as:*

$$v_i = \frac{1}{L-1} \sum_{k=1}^{L-1} \mathbf{1}\{\phi(s_k) = a, \phi(s_{k+1}) = b\} \quad (3.2)$$

where i is the index corresponding to the transition $a \rightarrow b$ in the order of the 16 possible transitions.

3.2 The 16 Polarity Interactions

The 16 possible polarity transitions are:

Table 3.1: The 16 polarity transitions with indices

Index	Transition	Index	Transition
0	$P^+ \rightarrow P^+$	8	$N \rightarrow P^+$
1	$P^+ \rightarrow P^-$	9	$N \rightarrow P^-$
2	$P^+ \rightarrow N$	10	$N \rightarrow N$
3	$P^+ \rightarrow NP$	11	$N \rightarrow NP$
4	$P^- \rightarrow P^+$	12	$NP \rightarrow P^+$
5	$P^- \rightarrow P^-$	13	$NP \rightarrow P^-$
6	$P^- \rightarrow N$	14	$NP \rightarrow N$
7	$P^- \rightarrow NP$	15	$NP \rightarrow NP$

3.3 The Rényi Entropy Suite

The Rényi entropy is a generalization of the Shannon entropy that introduces a parameter α that controls the sensitivity to different parts of the distribution. This family includes as special cases:

- $\alpha = 1$: Shannon entropy
- $\alpha = 2$: Collision entropy
- $\alpha \rightarrow \infty$: Min-entropy

Definition 3.3.1 (Rényi Entropy). *For a PIM vector $v \in \mathbb{R}^{16}$ and order $\alpha > 0$, $\alpha \neq 1$, the Rényi entropy is:*

$$H_\alpha(v) = \frac{1}{1-\alpha} \log_2 \left(\sum_i p_i^\alpha \right) \quad (3.3)$$

where $p_i = |v_i|/\|v\|_1$ is the probability of transition i .

For $\alpha = 1$, the entropy is the Shannon entropy:

$$H_1(v) = - \sum_i p_i \log_2(p_i) \quad (3.4)$$

For $\alpha \rightarrow \infty$, the entropy is the min-entropy:

$$H_\infty(v) = - \log_2(\max_i p_i) \quad (3.5)$$

THE 13 MATHEMATICAL METRICS

This chapter presents the complete mathematical description of all 13 metrics implemented in SGPMAIN (version 12.0). These metrics are derived from the PIM vectors and provide a comprehensive characterization of protein polarity profiles.

4.1 Shannon Entropy

The Shannon entropy measures the uncertainty or randomness in the polarity distribution.

Definition 4.1.1 (Shannon Entropy). *For a PIM vector $v \in \mathbb{R}^{16}$, the Shannon entropy is:*

$$H(v) = - \sum_i \frac{|v_i|}{\|v\|_1} \log_2 \left(\frac{|v_i|}{\|v\|_1} + \epsilon \right) \quad (4.1)$$

where $\|v\|_1 = \sum_i |v_i|$ is the L1 norm, and $\epsilon = 10^{-10}$ is a small constant to avoid division by zero. The entropy ranges from 0 to $\log_2(16) = 4$ bits.

4.2 Rényi Entropy ($\alpha = 2$ and $\alpha = 3$)

The Rényi entropy provides a generalized measure of uncertainty.

Definition 4.2.1 (Rényi Entropy). *For a PIM vector $v \in \mathbb{R}^{16}$ and order $\alpha > 0$, $\alpha \neq 1$, the Rényi entropy is:*

$$H_\alpha(v) = \frac{1}{1-\alpha} \log_2 \left(\sum_i p_i^\alpha \right) \quad (4.2)$$

where $p_i = |v_i|/\|v\|_1$ is the probability of transition i .

4.3 Gini Coefficient

The Gini coefficient measures the inequality in the polarity distribution.

Definition 4.3.1 (Gini Coefficient). *For a PIM vector $v \in \mathbb{R}^{16}$, the Gini coefficient is:*

$$G(v) = 1 - \frac{2}{n} \sum_{i=1}^n \frac{\sum_{j \leq i} p_{(j)}}{\sum_j p_j} \quad (4.3)$$

where $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(n)}$ are the sorted probabilities, and $n = 16$. The Gini coefficient ranges from 0 (perfect equality) to 1 (perfect inequality).

4.4 Grassmann Projection Distance

The Grassmann projection distance measures the distance between two subspaces in the Grassmann manifold.

Definition 4.4.1 (Grassmann Projection Distance). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Grassmann projection distance is:*

$$P_v = \frac{vv^T}{\|v\|^2}, \quad P_w = \frac{ww^T}{\|w\|^2} \quad (4.4)$$

$$d_{Gr}(v, w) = \frac{1}{\sqrt{2}} \|P_v - P_w\|_F \quad (4.5)$$

where $\|\cdot\|_F$ is the Frobenius norm. The distance ranges from 0 (identical subspaces) to 1 (orthogonal subspaces).

4.5 Fractal Dimension

The fractal dimension measures the self-similarity of the polarity profile.

Definition 4.5.1 (Fractal Dimension). *For a PIM vector $v \in \mathbb{R}^{16}$, the fractal dimension is computed by:*

1. Compute the absolute values: $a_i = |v_i|$
2. Compute the cumulative sum: $c_i = \sum_{j=1}^i a_j$
3. Normalize: $\hat{c}_i = (c_i - \min c) / (\max c - \min c)$
4. For each scale $s = 1, \dots, S$, compute box count: $N_s = \#\{\text{unique boxes}\}$

$$D_f(v) = -\frac{\log(N_s)}{\log(s)} \quad (4.6)$$

The fractal dimension ranges from 0 to 1.

4.6 Summary Table of Metrics

Table 4.1: Summary of all 13 mathematical metrics in SGPMAN (version 12.0)

Metric	Symbol	Range
Shannon Entropy	$H(v)$	$[0, 4]$ bits
Rényi Entropy ($\alpha = 2$)	$H_2(v)$	$[0, 4]$ bits
Rényi Entropy ($\alpha = 3$)	$H_3(v)$	$[0, 4]$ bits
Gini Coefficient	$G(v)$	$[0, 1]$
Grassmann Projection Distance	$d_{Gr}(v, w)$	$[0, 1]$
Fractal Dimension	$D_f(v)$	$[0, 1]$
Wasserstein Distance	$W_1(v, w)$	$[0, 16]$
Structural Complexity Index	$C(v)$	$[0, 1]$
Functional Modularity Index	$M(v)$	$[0, 1]$
Moran's I	$I(v)$	$[-1, 1]$
Total Correlation	$TC(v)$	$[0, \infty)$
Bhattacharyya Mean	$D_B(v, w)$	$[0, \infty)$
Radon Mean	$\mathcal{R}(v)$	$\mathbb{R}^{16}$

CLIFFORD ALGEBRA AND GRASSMANN ANALYSIS

This chapter provides a detailed mathematical description of the Clifford algebra and Grassmann analysis operations used in SGPMAN (version 12.0).

5.1 Clifford Algebra Fundamentals

Clifford algebra is a natural generalization of the inner product and exterior (wedge) product. In SGPMAN, we use Clifford algebra to decompose the relationship between two PIM vectors into scalar, vector, and bivector components.

Definition 5.1.1 (Geometric Product). *For two vectors $v, w \in \mathbb{R}^{16}$, the geometric product is:*

$$vw = v \cdot w + v \wedge w \quad (5.1)$$

where:

- $v \cdot w$ is the scalar product (inner product): $v \cdot w = \sum_i v_i w_i$
- $v \wedge w$ is the wedge product (exterior product): $(v \wedge w)_{ij} = v_i w_j - v_j w_i$

The wedge product produces a bivector, which is a 16×16 antisymmetric matrix.

5.2 The Corrected Wedge Product

The corrected wedge product is used to compute the wedge similarity between two PIM vectors.

Definition 5.2.1 (Corrected Wedge Product). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the corrected wedge product is:*

$$\omega_K(v, w) = v \wedge w - \frac{1}{16} \text{Tr}(v \wedge w) \quad (5.2)$$

where Tr is the trace operation.

5.3 The Wedge Similarity

The wedge similarity is the primary measure of similarity between two PIM vectors.

Definition 5.3.1 (Wedge Similarity). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the wedge similarity is:*

$$\text{WedgeSim}(v, w) = 1 - \frac{\|\omega_{\mathcal{K}}(v, w)\|_2}{\|v\|_2 \|w\|_2} \quad (5.3)$$

where $\|\cdot\|_2$ is the Euclidean norm.

The wedge similarity ranges from 0 to 1, where 1 indicates identical polarity profiles and 0 indicates orthogonal profiles.

5.4 The Rotor Similarity

The rotor similarity represents the optimal rotation aligning one PIM vector to another.

Definition 5.4.1 (Rotor Similarity). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the rotor similarity is:*

$$R(v, w) = \langle v, w \rangle + \|v \wedge w\|^2 \quad (5.4)$$

where:

- $\langle v, w \rangle$ is the inner product: $\langle v, w \rangle = v \cdot w$
- $\|v \wedge w\|^2$ is the squared Frobenius norm of the wedge product

5.5 Optimal Geodesic on Grassmann Manifold

The optimal geodesic between two points on the Grassmann manifold is computed using the SVD decomposition.

Definition 5.5.1 (Optimal Geodesic). *For two points $P_v = vv^T/\|v\|^2$ and $P_w = ww^T/\|w\|^2$ on the Grassmann manifold $Gr(k, n)$, the optimal geodesic is:*

$$P(t) = P_v \cos^2(\theta t) + P_w \sin^2(\theta t) + \text{cross terms} \quad (5.5)$$

where:

- $t \in [0, 1]$ is the parameter along the geodesic
- θ is the principal angle between the subspaces
- The cross terms are computed from the SVD of P_v and P_w

5.6 Hodge Complementarity

The Hodge complementarity measures the degree to which two PIM vectors are complementary.

Definition 5.6.1 (Hodge Complementarity). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Hodge complementarity is:*

$$H(v, w) = \|\star(v \wedge w)\|_2 \quad (5.6)$$

where $\star$ is the Hodge star operator, which maps a bivector to a vector in the dual space.

5.7 Multilevel Grassmann Analysis

SGPMAIN (version 12.0) performs Grassmann analysis at multiple levels, capturing geometric structure at different scales.

Definition 5.7.1 (Multilevel Grassmann Analysis). *For a PIM vector $v \in \mathbb{R}^{16}$ and level $k \in \{1, 2, 3\}$:*

$$v^{(k)} = \text{project}_k(v) \tag{5.7}$$

where project_k projects the vector onto a k -dimensional subspace.

The analysis computes:

- *The wedge similarity at each level: $\text{WedgeSim}_k(v, w) = \text{WedgeSim}(v^{(k)}, w^{(k)})$*
- *The rotor similarity at each level: $R_k(v, w) = R(v^{(k)}, w^{(k)})$*
- *The geodesic distance at each level: $d_k(v, w) = d_{Gr}(v^{(k)}, w^{(k)})$*

Part III

Case Study: Ebola Virus (EBOV)

APPLICATION TO EBOLA VIRUS

This chapter demonstrates the application of SGPMAN (version 12.0) to the Ebola Virus (EBOV) glycoprotein. The case study shows how the system processes sequences, computes metrics, and designs therapeutic peptides, with specific reference to the *config_EBOLA.json* parameters used.

6.1 Introduction to Ebola Virus

The Ebola virus (EBOV) is a highly pathogenic virus belonging to the family Filoviridae. It causes severe hemorrhagic fever with high mortality rates in humans. The virus encodes a single glycoprotein (GP) that is responsible for receptor binding and membrane fusion, making it the primary target for therapeutic intervention.

The GP is a type I transmembrane protein that forms trimers on the viral surface. It consists of two subunits: GP1 (receptor binding) and GP2 (fusion). The GP is heavily glycosylated, which helps the virus evade the host immune system. The fusion loop, a highly conserved region in GP2, plays a critical role in the membrane fusion process.

Ebola Virus Species

Six species of Ebola virus have been identified:

- **Zaire ebolavirus (EBOV):** The most virulent species, responsible for the 2014-2016 West African outbreak.
- **Sudan ebolavirus (SUDV):** The second most virulent species, with a case fatality rate of around 50%.
- **Reston ebolavirus (RESTV):** Found in the Philippines, not known to cause disease in humans.
- **Bundibugyo ebolavirus (BDBV):** Identified in Uganda in 2007, with a case fatality rate of around 40%.
- **Tai Forest ebolavirus (TAFV):** Identified in Ivory Coast in 1994, with only one known human case.
- **Bombali ebolavirus (BOMV):** Identified in bats in Sierra Leone in 2018, not yet associated with human disease.

This case study focuses on the Zaire ebolavirus species, but the analysis also includes the other five species for comparison.

6.2 Data Loading and Processing Summary

The system successfully loaded and processed all protein sequences:

Table 6.1: Global processing summary for SGPMAIN (version 12.0) analysis

Metric	Value
Total sequences read	1,832,319
Total valid PIMs	1,832,319
Total rejected	0
Validity rate	100.00%
Total bytes processed	0.57 GB
Processing time	6 minutes 37 seconds
Average speed	4,613 sequences per second
Peak memory usage (streaming)	~8 MB
Peak memory usage (analysis)	~500 MB
Hash index buckets	1,792
Reference group used	EBOLA_ZAIRE

6.3 EBOV Glycoprotein: Sequence Information

The Ebola virus glycoprotein (GP) is a 676-residue protein. The complete sequence is:

```

MGVTGILQLPRDRFKRTSFFLWVILFQRTFSIPLGVIHNS
TLQVNDVDKLVCRDKLSSTNQLRSVGLNLEGNVATDVPSA
TKRWGFRSGVPPKVVNYEAGEWAENCYNLEIKKPDGSECLP
AAPDGIRGFPRCRYVHKVSGTGPCAGDFAFHKEGAFFLYDR
LASTVIYRGTTFAEGVVAFLILPQAKKDFSSHPLREPVNA
TEDPSSGGYSTTIRYQATGFGTNETEYLFVDNLTYVQLES
RFTPQFLLQLNETIYTSKRSNTTGKLIWKNPEIDTTIGE
WAFWETKKNLTRKIRSEELSFTVVSNGAKNISGQSPARTSS
DPGTNTTTEDHKIMASENSSAMVQVHSQGREAAVSHLTTLA
TISTSPQSLTTKPGPDNSTHNTPVYKLDISEATQVEQHHR
TDNDSTASDTPSATTAAAGPPKAENTNTSKSTDFLDPATTT
PQNHSETAGNNNTHHQDTGEESASSGKLGLITNTIAGVAGL
ITGGRRTTRREAIVNAQPKCNPNLHYWTTQDEGAAIGLAWIP
YFGPAAEGIIYTEGLMHNQNGLICGLRQLANETTQALQLFLR
ATTELRTFSILNRKAIDFLLQRWGGTCHILGPDCCIEPHDW
TKNITDKIDQIIHDFVDKTLDPDQGDNDNWWTGWRQWIPAGI
GVTGVIIAVALFCICKFVF

```

6.4 Key Properties of EBOV GP (Zaire)

The SGPMAIN system computed a comprehensive set of properties for the EBOV GP:

Table 6.2: Key properties of EBOV GP (Zaire) with 13 metrics

Property	Value
Length	676 amino acids
Function	Receptor attachment, membrane fusion
Shannon Entropy	3.4293 bits
Rényi Entropy ($\alpha = 2$)	3.0665 bits
Rényi Entropy ($\alpha = 3$)	2.8472 bits
Gini Coefficient	0.4191
Fractal Dimension	0.9624
Structural Complexity	1.7875
Functional Modularity	0.5809
Moran's I	-0.0667
Total Correlation	0.0
Bhattacharyya Mean	-4.07e-08
Radon Mean	0.9999999999
Structural Stability	1.0000
Drug-Likeness	0.8457
Activity Score	0.7000

6.5 Chemical Profiles of EBOV Species

The system generated detailed chemical profiles for each of the six Ebola virus species. The profiles include properties such as Net Charge, Charge Density, Charge Balance, GRAVY, Hydrophobicity Score, Hydrophobic Components, Solubility, ΔG , Tm, Stability Score, Hotspot analysis, Membrane Permeability, Membrane Affinity, and Buffer Recommendation.

Table 6.3: Selected chemical properties for EBOV species

Property	Zaire	Sudan	Reston	Bundibugyo	Tai For- est	Bombali
Net Charge	0.032	0.029	0.017	0.052	0.042	0.055
Charge Density	0.057	0.050	0.039	0.071	0.065	0.076
Charge Bal- ance	0.784	0.787	0.725	0.866	0.819	0.858
GRAVY	-0.260	-0.124	-0.030	-0.051	0.141	0.208
Hydrophobicity Score	0.161	0.172	0.182	0.175	0.186	0.196
Solubility (mg/mL)	13.41	13.12	12.87	13.18	12.81	12.81
ΔG (kcal/mol)	-7.22	-7.18	-7.16	-7.09	-7.07	-6.82
Tm ($^{\circ}\text{C}$)	42.89	42.72	42.63	42.37	42.29	41.27
Stability Score	0.645	0.636	0.632	0.618	0.614	0.563

Interpretation of Chemical Profiles

The chemical profiles reveal important differences between the Ebola virus species:

- **Net Charge:** The Bombali species has the highest net positive charge (0.055), while the Reston species has the lowest (0.017). This variation may affect the electrostatic interactions with the host cell receptor.
- **GRAVY:** The Zaire species is the most hydrophilic (GRAVY = -0.260), while the Bombali species is the most hydrophobic (GRAVY = 0.208). This may affect the membrane affinity and fusion properties.
- **Stability Score:** The Zaire species has the highest stability score (0.645), while the Bombali species has the lowest (0.563). This may correlate with the higher virulence of the Zaire species.
- **Hydrophobic Patch 1:** All species have a large hydrophobic patch (size 5) at positions 11-15, which is likely the fusion loop. This is a critical target for therapeutic intervention.
- **Hotspots:** All species have a hydrophobic hotspot and a charge-rich region, suggesting conserved functional domains that could be targeted by drugs.

6.6 PIDP Analysis for EBOV Species

The PIDP analysis provides critical insight into the structural flexibility of the Ebola virus glycoproteins:

Table 6.4: PIDP results for EBOV species

Species	Tool	Disorder (0.5)	Mean Score
Zaire	metapredict	39.94%	0.441
Zaire	AIUPred	38.91%	0.415
Sudan	metapredict	37.87%	0.429
Sudan	AIUPred	39.64%	0.423
Reston	metapredict	36.78%	0.423
Reston	AIUPred	41.06%	0.435
Bundibugyo	metapredict	40.38%	0.424
Bundibugyo	AIUPred	39.50%	0.421
Tai Forest	metapredict	38.31%	0.433
Tai Forest	AIUPred	41.57%	0.432
Bombali	metapredict	35.12%	0.407
Bombali	AIUPred	41.82%	0.443

Interpretation:

- All EBOV species show a significant amount of disorder (35-42% at threshold 0.5), which is consistent with the high flexibility required for the glycoprotein's function.
- The high disorder content is likely related to the GP's role in receptor binding and membrane fusion, which requires conformational changes.
- The Bombali species shows the lowest disorder (35.12%) according to metapredict, which may be related to its lower pathogenicity in humans.

6.7 Similarity Analysis: EBOV vs All Groups

The SGPMAIN (version 12.0) system compared the EBOV GP (Zaire) against all reference groups using the wedge similarity and rotor similarity metrics.

Table 6.5: Top 10 ranked comparisons of EBOV GP (Zaire) against all groups

Rank	Group	Wedge Sim.	Rotor Sim.
1	EBOLA_SUDAN	0.9785	0.9980
2	EBOLA_BUNDIBUGYO	0.9682	0.9960
3	EBOLA_TAI_FOREST	0.9677	0.9959
4	EBOLA_RESTON	0.9740	0.9973
5	EBOLA_BOMBALI	0.9214	0.9784
6	LCMV	0.9631	0.9948
7	NILE2	0.9402	0.9878
8	RVF4 (Gc-strain)	0.9696	0.9962
9	UNFOLDED	0.8642	0.9369
10	NON_CPP	0.8827	0.9474

Key Observations:

- The highest similarities are between different Ebola species, as expected.
- The Sudan species shows the highest similarity to Zaire (0.9785).
- The similarities to other viruses (LCMV, NILE2, RVF4) are also relatively high, suggesting some functional similarity.
- The rotor similarities are generally higher than the wedge similarities, indicating that the profiles can be partially aligned.

6.8 Similarity Matrix Analysis

The similarity matrix provides a comprehensive view of the relationships between all groups.

Table 6.6: Selected wedge similarities between EBOV species and other groups

Group	Zaire	Sudan	Reston	Bundibugyo	Tai Forest	Bombali
Zaire	1.000	0.979	0.974	0.968	0.968	0.921
Sudan	0.979	1.000	0.972	0.961	0.969	0.925
Reston	0.974	0.972	1.000	0.954	0.966	0.914
Bundibugyo	0.968	0.961	0.954	1.000	0.977	0.956
Tai Forest	0.968	0.969	0.966	0.977	1.000	0.947
Bombali	0.921	0.925	0.914	0.956	0.947	1.000

Key Observations:

- The Zaire and Sudan species are the most similar (0.979).

- The Bombali species is the most distinct, with the lowest similarities to the other species.
- The Tai Forest and Bundibugyo species are also highly similar (0.977).

6.9 Chemical Hotspot Analysis

The system identified drug-binding hotspots in the EBOV GP, based on the chemical profile.

Table 6.7: Hotspots identified in EBOV GP (Zaire)

Hotspot Name	Score	Key Components	Type
Hydrophobic Patch 1	0.3208	Components 11-15 (size 5)	Hydrophobic
Charge-rich Region	0.1075	Residues D, E, K, R	Charged
Hydrophobic Patch 2	0.0244	Component 3 (size 1)	Hydrophobic

Interpretation:

- **Hydrophobic Patch 1:** This is the most significant hotspot, with a high score of 0.3208. It corresponds to the fusion loop, a highly conserved region in the GP2 subunit. This is a critical target for therapeutic peptides that aim to inhibit membrane fusion.
- **Charge-rich Region:** This hotspot (score 0.1075) corresponds to a cluster of charged residues (D, E, K, R) that may be involved in receptor binding or electrostatic interactions with the host cell.
- **Hydrophobic Patch 2:** This is a smaller hydrophobic hotspot (score 0.0244) that may represent a secondary binding site or a region involved in GP trimerization.

6.10 Cross-Strain Similarity Matrix (Table 6)

The cross-strain similarity matrix quantifies the functional relationships among all six Ebola glycoprotein strains using SGPMAN12.0v wedge similarity values.

Table 6.8: Cross-strain similarity matrix for Ebola glycoproteins (SGPMAN12.0v).

Strain	Sudan	Zaire	Reston	Bundibugyo	Tai Forest	Bombali
Sudan	1.000	0.462	0.461	0.480	0.480	0.47
Zaire	0.462	1.000	0.470	0.490	0.489	0.48
Reston	0.461	0.470	1.000	0.488	0.488	0.47
Bundibugyo	0.480	0.490	0.488	1.000	0.509	0.49
Tai Forest	0.480	0.489	0.488	0.509	1.000	0.50
Bombali	0.47	0.48	0.47	0.49	0.50	1.000

Key Observations:

- Similarity values range from 0.461 to 0.509, revealing moderate functional similarity across all strains.
- Tai Forest and Bundibugyo are the most similar (0.509).
- Sudan and Reston are the least similar (0.461).
- Bombali shows intermediate similarity to all other strains (0.47-0.50).

6.11 Comparative Analysis: SGPM12.0v vs BioPython (Table 7)

The comparative analysis highlights the superior discriminative power of SGPM12.0v against BioPython across selected reference groups.

Table 6.9: Comparative analysis of SGPM12.0v and BioPython for selected reference groups.

Reference Group	SGPM12.0v	BioPython	Interpretation
LASV	0.811-0.891	0.000	SGPM12.0v detects high similarity
LCMV	0.836-0.866	0.000-0.008	BioPython fails to detect similarity
VIRUS_REVIEWED	0.749-0.817	0.011-0.068	Both methods show moderate similarity
VIRUS_UNREVIEWED	0.787-0.854	0.000-0.036	Only SGPM12.0v detects similarity
CPP	0.231-0.270	0.052-0.081	Both methods show low similarity
NON_CPP	0.567-0.612	0.105-0.192	Different sensitivity levels
UNFOLDED	0.495-0.531	0.060-0.088	Both methods show low similarity
PARTIALLY_FOLDED	0.480-0.503	0.012-0.056	Both methods show low similarity

Key Observations:

- SGPM12.0v demonstrates superior discriminative power, detecting high similarity with LASV and LCMV.
- BioPython returns values close to zero for LASV and LCMV, failing to detect functional similarity.

- Both methods show low similarity with CPP, confirming Ebola glycoproteins are functionally distinct from cell-penetrating peptides.

6.12 Peptide Optimization Results

Base Peptide Evaluation

The base peptide sequence (loaded from *config_EBOLA.json*) is a known inhibitor of the Ebola virus glycoprotein. The system evaluated the base peptide using all 13 metrics.

Table 6.10: Initial evaluation of base peptide

Metric	Value
PIM Similarity	0.000
Antiviral Activity	0.700
Drug Score	0.846
Structural Stability	1.000
Grassmann Distance	0.000
Hodge Complementarity	0.0759
Entropy	3.4293
Rényi Entropy ($\alpha = 2$)	3.0665
Gini Coefficient	0.4191
Fractal Dimension	0.9624
Ricci Curvature	563232909.5101
Drug-Likeness	0.8457
Recommendation	MODERATE - Optimize before validation

Optimized Peptide

The optimization process identified an improved candidate peptide.

Table 6.11: Comparison of base and optimized peptide properties

Metric	Base Peptide	Optimized Peptide
Antiviral Activity	0.700	0.9185
Drug Score	0.846	0.985
PIM Similarity	0.000	0.882
Grassmann Distance	0.000	0.438
Hodge Complementarity	0.0759	0.044
Structural Stability	1.000	0.982
Entropy	3.4293	3.124
Rényi Entropy ($\alpha = 2$)	3.0665	2.847
Fractal Dimension	0.9624	0.612
Drug-Likeness	0.8457	0.731
Recommendation	MODERATE	EXCELLENT

Improvements:

- Activity: +31.2% (0.700 $\rightarrow$ 0.9185)
- Drug Score: +16.4% (0.846 $\rightarrow$ 0.985)
- PIM Similarity: Significant improvement
- Grassmann Distance: Significant improvement

6.13 Therapeutic Profile Summary

The system generated a comprehensive therapeutic profile for the EBOV GP.

Table 6.12: Therapeutic profile for EBOV GP

Property	Value
Peptide Sequence	TFLHREWFMTTNLTWSSAGSTVWR
Length	25 amino acids
Predicted Antiviral Activity	0.9185
Predicted Drug Score	0.985
Structural Stability	0.982
Interaction Quality	0.044
Structural Complementarity	Low
Recommendation	EXCELLENT - Proceed to in vitro validation

6.14 Biochemical Recommendations

Based on the peptide's predicted properties, the following recommendations were generated:

1. **SYNTHESIZE:** Sequence TFLHREWFMTTNLTWSSAGSTVWR by solid-phase synthesis
2. **FORMULATE:** PBS pH 7.4 buffer
3. **PROTECT:** Add protecting groups at N and C termini
4. **VALIDATE:** GP binding assays (SPR/ITC)
5. **OPTIMIZE:** Consider conformational constraints to increase sequence diversity
6. **STRUCTURAL ANALYSIS:** Perform CD spectroscopy to confirm secondary structure
7. **STABILITY TESTING:** Evaluate plasma stability and proteolytic resistance

INSTALLATION AND SYSTEM REQUIREMENTS

This appendix provides concise information on system requirements, installation steps, and how to run SGPMAN (version 12.0). It also includes the execution log summary from the September 2026 run and details about the mathematical improvements.

A.1 System Requirements

Table A.1: Minimum system requirements for SGPMAN (version 12.0)

Component	Requirement
Operating System	Linux, macOS, or Windows (with WSL)
Python	Version 3.8 or higher
RAM	8 GB (32 GB recommended for ESM2)
Storage	20 GB free space
Processor	64-bit, 4 or more cores
GPU (optional)	NVIDIA GPU with 4+ GB VRAM

A.2 Required Packages

```
numpy pandas scipy scikit-learn matplotlib seaborn psutil
torch transformers peft tokenizers accelerate
metapredict aiupred (optional for PIDP)
```

A.3 Installation Steps

1. Clone the repository:

```
git clone https://github.com/polancodf2024/ebov-sgp-analysis.git
cd ebov-sgp-analysis
```

2. Install core dependencies:

```
pip install numpy pandas scipy scikit-learn matplotlib seaborn psutil
```

3. Install deep learning dependencies:

```
pip install torch transformers peft tokenizers accelerate
```

4. (Optional) Install PIDP dependencies:

```
pip install metapredict aiupred
```

A.4 Configuration File Setup

The `config_EBOLA.json` file must be placed in the working directory. The system will automatically detect and load it. If `config_EBOLA.json` is not present, the system falls back to hardcoded default values.

Minimum structure of config.json for version 12.0:

```
{
  "metadata": {
    "version": "1.0.0",
    "schema_version": "2.0.0",
    "sgpmain_min_version": "12.0.0"
  },
  "metric_weights": {
    "pim": 0.28, "entropy": 0.12, "grassmann": 0.14,
    "hodge": 0.08, "curvature": 0.06, "gini": 0.04,
    "fubini": 0.04, "jensen_shannon": 0.06, "spearman": 0.04,
    "hellinger": 0.04, "wasserstein": 0.04, "fractal": 0.03, "radon": 0.03
  },
  "target_ranges": {
    "ebola_zaire": {
      "pim_similarity": {"min": 0.80, "max": 0.95},
      "entropy_shannon": {"min": 3.0, "max": 4.0},
      "entropy_renyi_alpha2": {"min": 2.5, "max": 3.5},
      "grassmann_distance": {"min": 0.20, "max": 0.40}
    }
  },
  "base_peptide_sequence": {
    "ebola_zaire": {
      "sequence": "MGVTGILQLPRDRFKRTSFF...",
      "peptide_name": "EBOV-GP_inhibitor",
      "length": 676
    }
  },
  "processing_params": {
    "batch_size": 5000,
```

```
        "max_stored_proteins": 200,  
        "n_bootstrap": 50,  
        "max_workers": 4  
    }  
}
```

A.5 Running SGPMAN (version 12.0)

1. Place your FASTA files in the working directory:

```
zaire.unico.dat0  sudan.unico.dat0  reston.unico.dat0  ...
```

2. (Optional) Place config.json in the working directory.
3. Run the program:

```
python3 SGPMAN212.0v.py
```

4. Results are saved in:

```
results_characterization_YYYYMMDD_HHMMSS/
```

A.6 Quick Reference

```
# Run full analysis  
python3 SGPMAN212.0v.py
```

```
# View results  
ls results_characterization_*/
```

```
# View summary  
cat results_characterization_*/multidisciplinary_summary_table.txt
```


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